Computing and Software

SFWRENG 2DM3

Discrete Mathematics with Applications I

Fall 2024

Instructor Information

Ridha Khedri

Email: khedri@mcmaster.ca

Office: ITB-129

Office Hours:

Thursday: 3:00 PM to 4:00PM

I will be available at my office ITB-129

TA Information

Name: Ruidi Liu

Email: liu127@mcmaster.ca

Name: Lucas Dutton

Email: duttonl@mcmaster.ca

Name: Hamza Jamil

Email: jamilh12@mcmaster.ca

Name: Michael Roberts

Email: roberm67@mcmaster.ca

Class Times

Lectures: Three lectures per week

Labs: None

Tutorial Material:

In most weeks, exercises will be provided, from which the main material for the tutorials will be taken. Every week, starting September 09, there will be one-hour tutorial session in four tutorial groups. The main purpose of the tutorials is to discuss student work on exercise problems. Therefore, every student is expected to complete the scheduled work, i.e., exercise problems or necessary reading, before the corresponding tutorial session. In particular, solutions and solution attempts to the Exercises of the current week are to be brought to the tutorial.

Class Format

In Person

Course Dates: 09/03/2024 - 12/05/2024

Units: 3.00

Course Delivery Mode: In Person

Course Description: Functions, relations and sets; the language of predicate logic, propositional logic; proof techniques, counting principles; induction and recursion, discrete probabilities, graphs, and their application to computing. Three lectures, one tutorial (one hour); first term Prerequisite(s): MATH 1ZC3; registration in any Software Engineering or Mechatronics Engineering program Antirequisite(s): COMPSCI 2DM3, SFWRENG 2E03, 2F03

Instructor-Specific Course Information

This course will teach the mathematical foundations of software engineering. The course general objective is to provide students with an overview of discrete mathematics. It includes mathematical knowledge and skills concerning syntax, semantics, pragmatics, and vocabulary of the language of (discrete) mathematics. Conscious and precise use of this language is the foundation for precise specification and rigorous reasoning, which take a central place in this course.

Course Precondition

Students are expected to have achieved the following learning objectives before taking this course:

1. Students should know and understand

- (a) Basic number types, such as integers, rationals, reals
- (b) General principles of mathematical notation
- (c) General principles of calculation in mathematics
- (d) Basic imperative programming

2. Students should be able to

- (a) Perform calculations and solve equations in numerical domains
- (b) Write and "mentally execute" simple imperative programs

Course Postcondition

Students are expected to achieve the following learning objectives as a result of taking this course:

1. Students should know and understand

- (a) Syntax of propositional logic
- (b) Syntax of predicate logic, including variable binding issues
- (c) Principles of typed expressions, and the types of the operators they are using
- (d) Basic semantics of expressions and formulae (truth tables, validity, . . .)
- (e) Principles of calculational proofs and traditional-style (informal) proofs
- (f) Basic and derived proof rules of propositional logic
- (g) Basics of (typed) set theory
- (h) Basic concepts and theorems about functions and (especially binary) relations
- (i) Basics of program specification using pre- and post-conditions
- (j) Basic theory of integers and counting
- (k) Basic graph theory

2. Students should be able to

- (a) Extract the propositional-logic structure from English sentences, and translate propositional expressions back into English
- (b) Translate English statements of moderate complexity into predicate-logic specification
- (c) Translate simple mathematical prose into predicate-logic definitions

- (d) Produce proofs in propositional logic
- (e) Explain and prove basic properties using induction
- (f) Produce correctness proofs for simple imperative programs
- (g) Produce proofs in applications to set and relation theory, integers and counting, etc.
- (h) Explain and prove basic graph properties.

Meeting Details

The lectures and tutorials are all in-person. During the office hour, the instructor will be available in ITB-129 (office). He can be contacted in-person or via Microsoft Teams.

Important Links

- Mosaic
- Avenue to Learn
- Student Accessibility Services - Accommodations
- McMaster University Library
- eReserves

It is the student's responsibility to be aware of the information on the course's Avenue to Learn page and to check regularly for announcements.

The latest version of the slides used in class, the material related to the course assignments, the material needed or used in the tutorial sessions, and announcements at the course website on Avenue to Learn.

Course Learning Outcomes

Logical Formalism

Calculational Proofs in Propositional Logic

Calculational Proofs in Predicate Logic Applications

Formalisation

Discrete Structures: Sets, Functions, Relations, Graphs

Discrete Probabilities and counting

Graduate Attributes

The Canadian Engineering Accreditation Board (CEAB) is a division of Engineers Canada and is responsible for accrediting undergraduate engineering programs across Canada. Accreditation by the CEAB ensures that the engineering programs meet a national standard of quality and cover essential educational requirements. Graduate Attributes are a set of qualities and skills that the CEAB expects engineering graduates to possess. These attributes are a benchmark for the learning outcomes of accredited engineering programs. This section lists the Graduate Attribute Indicators associated with the Learning Outcomes in this course.

Lab Safety

The Faculty of Engineering is committed to McMaster University's Workplace and Environmental Health and Safety Policy which states: "Students are required by University policy to comply with all University health, safety and environmental programs". It is your responsibility to understand McMaster University Workplace and Environmental Health and Safety programs and policies. For information on these programs and policies please refer to McMaster University Health and Safety. The Lab Safety Handbook is available here, as well as on A2L.

It is also your responsibility to follow any specific Standard Operating Procedures (SOPs) provided for some of the experiments and the laboratory equipment. A laboratory-specific set of rules can also be added to ensure that students fully understand laboratory safety rules that are in place prior to their first session.

Course Schedule

Topic	Chapters covered from textbook	Duration
Preliminaries –What is Discrete Mathematics, Elementary notions, Introduction to Calculational Reasoning–	Parts of Chapters 1, 15	≈ 1 week

Boolean Expressions and Propositional Logic	Chapters 1– 5	≈ 2 weeks
Sets and operation on them, ordered sets, finite versus infinite sets, and counting finite sets	Chapters 11 and 20	≈ 1.5 week
Quantification and Predicate Logic	Chapters 8–9	≈ 2 weeks
Relations and Functions	Chapter 14	≈ 2 weeks
Predicates and Programming	Chapter 10	≈ 1 week
Induction and Sequences	Chapters 12–13	≈ 1 week
Discrete probabilities and counting	Chapter 16	≈ 1.5 week
Graphs	Chapter 19	≈ 1 week

Required Materials and Texts

Textbook Listing: https://textbooks.mcmaster.ca

LOGICAL APPROACH TO DISCRETE MATH

ISBN: 3-540-94115-0

Authors: David Gries and Fred B. Schneider

Publisher: Springer

The textbook is availble at McMaster Bookstore.

Course Evaluation

Assignments

There will be four (4) assignments consisting of problems requiring mathematical constructions and proofs. It is **desirable** to use LaTeX documentation preparation system to write your solutions to the assignments. If you use LaTeX, you must submit the PDF file. We also accept (readable) hand-write or tablet-write submitted pdf files.

The assignments will be marked by the TAs. Some assignments may contain bonus questions. All bonus marks will be added to the course grade only for those who have passed the course otherwise.

The following table gives a tentative timetable for the assignments.

Assignment set	**Handed out date**	**Due date**	**Weight**
A1	September 12	September 23	5%
A2	September 26	October 14	5%
A3	October 31	November 18	5%
A4	November 21	December 04	5%

Assignments, tests and the final exam may be graded only automatically. It is essential that you meet the deadlines for the assignments; there is no credit for material submitted after the deadline.

For each student, **only the three (3) best grades**, among the four grades for the assignments, will be counted towards the final grade. Hence, <u>the overall weight of the assignments is 15%.</u>

Midterms

On Monday, October 21, there will be one 50-minute midterm test during the normal lecture time. <u>The mid-term exam is worth 30% of the course final grade.</u>

Final Exam

The final exam will be 2 hours and 30 minutes long. It will test accumulative knowledge with more emphasis on the recent material not covered by the midterms. It will take place on the date scheduled by the University. The final exam counts for 55% of your course final grade.

IMPORTANT NOTE:

To pass the course, you must gain a grade of at least 50% on the midterm and the final exam, or on the final exam only. In other terms, you must have the TWO following conditions satisfied (Satisfying only one of the two below conditions is not sufficient to pass the course):

1. **Max(((Final_Exam_Grade/55)*100), (((Final_Exam_Grade + Midterm_Exam_Grade)/85)*100)) >= 50**
2. **Final_Exam_Grade + Midterm_Exam_Grade + The grades of the three best assignments > = 50**

The midterm exam will be marked over 30 points and the final exam will be marked over 55 points.

If you pass the course (i.e., your grades satisfy the above conditions 1 AND 2), the course final grade is calculated as follows:

Course_Final_Grade = Final_Exam_Grade + Midterm_Exam_Grade + (The grades of the three best assignments).

The instructor reserves the right to conduct any deferred midterm orally.

Course Evaluation Details

Format for submission:

- PDF (single file)
- You can hand-write, tablet-write, or use LaTeX.
- If you hand-write (or table-write), please use a white (lined is ok) background, with dark blue or black writing. If you use pencil, it needs to be dark.
- Only write on one side of the paper (if you write on both, both sides show up in your scan!)
- If you're scanning it in, please use high (300 DPI) resolution.

Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Late Assignments

It is essential that you meet the deadlines for the assignments; there is no credit for material submitted after the deadline.

Absences, Missed Work, Illness

1. Missed deliverable or exam: A zero will be entered for the missed work unless the instructor receives a notification from the Associate Dean office or an MSAF within a reasonable time. It is YOUR responsibility to follow up with your instructor immediately (normally within 2 working days) to discuss possible consideration.
2. Tutorials: The class will be split into tutorial groups. Each group will have a one hour tutorial per week. Attendance at tutorials is compulsory. You MUST attend your assigned session of the tutorial. More information on the schedule and the organization of the tutorial will be posted on the course

avenue page.

3. The instructor reserves the right to adjust the grades for an assignment, midterm exam, or the final exam by decreasing every score by a fixed number of points.
4. The instructor reserves the right to increase by a fixed number of points the final scores of all the students who handed in all the deliverables and attended all the tutorial sessions (except if the student misses a tutorial for a justifiable reason).
5. The instructor reserves the right to assign extra grades for extra work done by willing students. However, the work subject to extra grades should be advertised during the lectures.
6. A remarking request of an assignment or exam is considered by the TAs and the instructor **only if it is made within the two weeks that follow the return date of the majority of the concerned assignment or exam.**
7. The instructor reserves the right to require a deferred exam to be oral.
8. No responsibility for loss of assignment copy can be assumed by either the instructor or the Teaching Assistants. Keep copies of your submitted assignments work.
9. Calculators are not needed for this course and their use will not be permitted during exams.
10. The lectures will not necessary follow the order in which the topics are presented in the detailed course outline. Regular class attendance is required.
11. Significant study, reading, and group discussions outside of class and tutorials are required.
12. Suggestions on how to improve the course and the instructor's teaching methods are always welcomed.

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with McMaster academic integrity policy, it "shall be an offense knowingly to … submit academic work for assessment that was purchased or acquired from another source". This includes work created by generative AI tools. Also state in the policy is the following, "Contract Cheating is the act of "outsourcing of student work to third parties" (Lancaster & Clarke, 2016, p. 639) with or without payment." Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

Generative AI: Some Use Permitted

Example One

Students may use generative AI in this course in accordance with the guidelines outlined for each assessment, and so long as the use of generative AI is referenced and cited following citation instructions given in the syllabus. Use of generative AI outside assessment guidelines or without citation will constitute academic dishonesty. It is the student's responsibility to be clear on the limitations for use for each assessment and to be clear on the expectations for citation and reference and to do so appropriately.

Example Two

Students may use generative AI for [editing/translating/outlining/brainstorming/revising/etc] their work throughout the course so long as the use of generative AI is referenced and cited following citation instructions given in the syllabus. Use of generative AI outside the stated use of [editing/translating/outlining/brainstorming/revising/etc] without citation will constitute academic dishonesty. It is the student's responsibility to be clear on the limitations for use and to be clear on the expectations for citation and reference and to do so appropriately.

Example Three

Students may freely use generative AI in this course so long as the use of generative AI is referenced and cited following citation instructions given in the syllabus. Use of generative AI outside assessment guidelines or without citation will constitute academic dishonesty. It is the student's responsibility to be clear on the expectations for citation and reference and to do so appropriately.

Generative AI: Unrestricted Use

Students may use generative AI throughout this course in whatever way enhances their learning; no special documentation or citation is required.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

Equity, Diversity, and Inclusion

The Faculty of Engineering is committed to creating an environment in which students of all genders, cultures, ethnicities, races, sexual orientations, abilities, and socioeconomic backgrounds have equal access to education and are welcomed and treated fairly. If you have any concerns regarding inclusion in our Faculty, in particular if you or one of your peers is experiencing harassment or discrimination, you are encouraged to contact the Chair, Associate Undergraduate Chair, Academic Advisor or to contact the Equity and Inclusion Office.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

Academic Advising

For any academic inquires please reach out to the Office of the Associate Dean (Academic) in Engineering located in JHE-Hatch 301.

Details on academic supports and contact information are available from:

https://www.eng.mcmaster.ca/programs/academic-advising

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the Policy on Requests for Relief for Missed Academic Term Work.

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.